

Econ 646 Spring 2012
Planned Homework 2

This homework is due on Feb 22nd, at the beginning of the tutorial.

1. *King and Rebelo model*

- (a) Replicate Table 3 of King and Rebelo (2000)
- (b) Calculate the same table using the simulated data detrended with the first difference.

2. (The following questions are the continuation from the problem in the first homework.)

- (a) Next, we will explore an alternative fix. In this question and all the following questions, we will assume that $u(C_t, N_t) = \log C_t - \frac{\eta}{1+\eta} N_t^{\frac{\eta+1}{\eta}}$, but modify the definition of N_t to

$$N_t \equiv \left[N_{c,t}^{\frac{\theta+1}{\theta}} + N_{i,t}^{\frac{\theta+1}{\theta}} \right]^{\frac{\theta}{\theta+1}} \quad (1')$$

where $\theta \geq 0$. That is, N_t is now not a simple addition of sectoral labor input but is “produced” by the constant elasticity of substitution function (1’). Notice that (1’) is a generalization of $N_t = N_{c,t} + N_{i,t}$ since (1’) converges to $N_t = N_{c,t} + N_{i,t}$ when θ goes to infinity. Everything else is the same as before. Derive the first order conditions for the planner’s problem.

- (b) Show that C_t cancels out from the optimality condition $-(\partial u / \partial N_{c,t}) / (\partial u / \partial C_t) = (\partial C_t / \partial N_{c,t})$.
- (c) Comovement between N_t and $N_{c,t}$ is a necessary condition for the comovement between $N_{i,t}$ and $N_{c,t}$. Show that N_t and $N_{c,t}$ always move in the same direction if θ is smaller than some threshold value. Be specific to the threshold value in your proof.
- (d) Give some intuition to why small θ can help sectoral labor comovement.